

Consciousness as Field Resonance

From Ego-Filtered Ontology to Differentiation-Dominant Dynamics

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Abstract

*The ontology of consciousness remains constrained by a hidden assumption: that awareness is a local property generated inside isolated systems and accessed through the ego as observer. This paper argues that such an ontology is internally incomplete because the ego functions as a filter that cannot observe the conditions of its own operation. We propose that consciousness is best understood as a **resonant regime within a universal dynamical field**, where systems differ by their **capacity to form differentiated, self-referential dynamical structures**. Empirical analysis of EEG under propofol sedation (N=12, Chennu et al. 2016) shows spectral differentiation (Omega) tracks conscious state (Cohen's $d = 1.50$) with a double dissociation: identical drug concentrations produce opposite entropy changes. Global causal integration (kappa) does not capture an independent axis ($d = 0.42$); a pre-registered multiplicative model is falsified ($F1$). A regional posterior product ($d = 2.10$, exploratory) and negative Omega-kappa correlation ($r = -0.27$) suggest a **differentiation-dominant, regionally constrained resonance model**.*

Keywords: consciousness, spectral entropy, transfer entropy, differentiation, integration, field ontology, propofol, EEG, Energy-Flow Cosmology, Consciousness-Ego-Mirror

1. The Ontological Filter Problem

Most contemporary theories of consciousness assume three things: that consciousness is produced *inside* certain systems, that the ego is the observer of consciousness, and that measurement can access consciousness externally. These assumptions contain a structural blind spot:

The ego is part of the system it tries to observe. A filter cannot see the filtering conditions that define its own perception.

This circularity is not merely philosophical. When we classify a sedated subject as 'drowsy' because they score 3/40 on a behavioural task, we use the ego's output as a proxy for consciousness. But absence of ego-mediated response does not entail absence of experience — it entails only the filter's failure to produce measurable output. Patients in vegetative states have shown covert awareness through neuroimaging [5], demonstrating that the ego-filter can fail while experience persists.

1.1 The Three Standard Responses

Functionalism defines consciousness by what it does, but 'what it does' is observed through the ego-filter. **Integrated Information Theory** (IIT) defines consciousness as maximal integrated information, but the formalism was designed by an ego that already knows what consciousness feels like [8]. **Global Workspace Theory** identifies consciousness with global broadcast, but 'global' and 'broadcast' are spatial metaphors from the ego's representational framework. None of these acknowledge that the entity constructing the theory is part of the phenomenon being theorised.

2. A Field-Based Ontology

Level 1 — The Field. A universal dynamical background of energy flow and entropy dynamics in which all physical systems are embedded. This is not a 'consciousness field' in a panpsychist sense — it is the recognition that all matter participates in thermodynamic processes.

Level 2 — Resonance Capacity. Systems differ in their ability to generate rich, self-referential dynamical patterns within the field. A stone has minimal internal dynamical freedom. A waking human brain has extreme capacity. This is continuous, not binary.

This does **not** mean 'everything is conscious.' It means everything exists within the same dynamical substrate, but only some systems achieve the complexity required for experiential regimes.

2.1 Ego as Vector, Mirror as Operator

Within the Consciousness-Ego-Mirror (CEM) framework [3], the ego is a **vector component of energy flow**: direction, coherence, persistence. The **mirror** is the complementary operator: negative feedback that monitors how the ego's direction alters the surrounding field and returns corrective information. The ego is not the investigator of consciousness — it is one component of a triadic dynamical structure (field, vector, operator). The ontological filter problem dissolves: we recognise the ego as a pattern within the field that we can measure from within.

3. Mapping Ontology to Measurement

Ontological term	Dynamical correlate	Measurement
Field participation	Existence of measurable spectral dynamics	All EEG shows $\Omega > 0$
Ego (vector structure)	Internal differentiation / structured complexity	Spectral entropy (Ω)
Mirror (feedback regulator)	Regional causal closure	Transfer entropy (κ) in posterior cortex
Balance constraint	Trade-off between complexity and binding	$\text{corr}(\Omega, \kappa) < 0$

Table 1. Mapping between ontological concepts and empirical measures.

4. Empirical Results: TO-CS-02

4.1 Dataset

EEG data from Chennu et al. [1]: 20 healthy participants under graded propofol sedation. At moderate sedation (~ 1000 ug/L), subjects classified as responsive ($N=13$, $>24/40$) or drowsy ($N=7$, $<24/40$) despite equivalent drug concentrations. Definitive analysis uses the Overlap-12 cohort (7 responsive, 5 drowsy): Ω from validated power spectra and κ from raw time series, on the same individuals.

4.2 Differentiation (Ω)

Spectral entropy separates groups with **Cohen's $d = 1.50$** (full band) and shows a double dissociation: responsive subjects increase ($\Delta\Omega = +0.054$, $p = 0.001$), drowsy subjects decrease ($\Delta\Omega = -0.063$, $p = 0.016$). The drug does not determine direction — consciousness does. Centro-parietal cortex dominates ($d = 1.75$); occipital weakest ($d = 1.29$). Removing delta drops d to 0.56.

4.3 Integration (κ)

Global transfer entropy: **$d = 0.42$** , not significant. This does not prove integration is irrelevant — it shows that global TE as operationalised here ($k=3$, 6 bins, ROI-averaged) does not capture an independent axis. Alternative measures or finer resolution might reveal effects this analysis misses.

4.4 The Product Test: F1

Pre-registered primary test: does $C = \Omega(4-40 \text{ Hz}) \times \kappa$ beat $\Omega(4-40 \text{ Hz})$ alone? **No**. Bootstrap $\Delta d = -0.11$, $p = 0.52$. Score: **F1 — falsified**.

Measure	Cohen's d	p	Interpretation
Ω (full, 0.5-45 Hz)	1.50	0.073	Strong — primary marker
Ω (delta-free, 4-40 Hz)	0.56	0.202	Weak without delta
κ (global TE)	0.42	0.530	Weak, not significant
$C = \Omega(\text{full}) \times \kappa$	1.77	0.048	Driven by Ω
$C = \Omega(4-40) \times \kappa$	0.63	0.432	No improvement (F1)
$C(\text{parietal})$ — exploratory	2.10	0.018	Candidate for replication
$\text{corr}(\Omega, \kappa)$	-0.27	—	Trade-off

Table 2. All measures, Overlap-12 (7 responsive, 5 drowsy). Dissociation: $\text{corr}(\Omega, \kappa) = -0.267$.

4.5 Regional Feedback Signal

Parietal Omega x kappa achieves $d = 2.10$ — the largest observed effect. This was pre-registered as a secondary prediction, not a primary endpoint. It suggests a possible regional integration role requiring independent testing. It is a candidate hypothesis, not a confirmed effect.

4.6 The Balance Signal

$\text{corr}(\text{Omega}, \text{kappa}) = -0.27$: higher spectral entropy associates with lower transfer entropy. The conscious brain is not maximally coupled — it is maximally differentiated *within* a regionally sufficient integration structure.

5. The Revised Model

Falsified: Consciousness = Omega x kappa (both maximised globally)

Supported: Consciousness = differentiation-dominant resonance regime, constrained by sufficient regional feedback, in a balance regime.

5.1 Three Empirical Pillars

Pillar 1: Differentiation is primary. Omega $d = 1.50$, double dissociation, centro-parietal dominance.

Pillar 2: Global integration not detected; regional integration is a candidate. Global kappa $d = 0.42$. Parietal C $d = 2.10$ (exploratory). If confirmed, the CEM mirror-operator is anatomically localised [2].

Pillar 3: Balance, not maximisation. Negative Omega-kappa correlation ($r = -0.27$). Consistent with criticality and metastability models [6].

6. The Resonance Continuum

System	Field (Level 1)	Resonance (Level 2)	Empirical Omega
Stone	Yes (thermodynamic)	Near zero	Not measurable
Cell	Yes (metabolic)	Low	Metabolic entropy
Anaesthetised brain	Yes (neural)	Reduced	0.52 - 0.72
Waking human brain	Yes (neural)	Extreme	0.78 - 0.88

Table 3. The resonance continuum. Omega values from this study.

This is not panpsychism. A stone has minimal internal dynamical freedom, therefore almost no resonance structure. Even within human brains, the continuum is smooth: sub-08 (Omega = 0.52, 0/40) and sub-03 (Omega = 0.88, 33/40) occupy different positions on a continuous dimension.

7. Consciousness as Regime Cycle (Conceptual Extrapolation)

Note: This section presents theoretical extrapolations not directly tested by the EEG data reported here.

Phase	Dynamical regime	EFC description	Omega
Deep sleep	Low differentiation, high local sync	Minimal resonance	Low
REM / dreaming	High internal simulation	Internal resonance	Moderate
Awakening	Critical transition	Phase transition	Rising
Waking	High diff. + regional feedback	Full resonance	High
Fatigue	Entropy drift	Resonance decay	Declining
Sleep / death	Phase collapse	Exit from regime	Near zero

Table 4. Consciousness phases as dynamical regimes.

Each day, a biological system traverses a closed curve in state space: deep sleep through awakening into waking consciousness and back via fatigue to sleep onset. The same pattern extends to the life cycle. This suggests a structural isomorphism: in EFC cosmology, structure formation follows regime progression (L0-L3); in consciousness, the same dynamical topology applies. This is not identity — it is the same class of thermodynamic arc.

8. Ontological Implications

The field-resonance model addresses the filter problem by not requiring the ego to observe consciousness from outside. We measure spectral entropy independently of self-report. The double dissociation shows it tracks consciousness, not pharmacology.

This reframes the hard problem: not 'how does the ego produce consciousness?' but 'how does the field produce the ego?' — an empirically tractable question.

The field ontology (Level 1) is a theoretical interpretation consistent with the data but not directly tested by this experiment. The empirical contribution is the regime characterisation; the ontological framework organises interpretation but does not follow deductively from measurements.

9. Testable Predictions

ID	Prediction	Falsification test
R1	Posterior C(P_L) outperforms global C	DOC patients, sleep, ketamine
R2	kappa shows threshold/sigmoid relation	kappa vs hit-rate; linear wins = fails
R3	Negative corr(Omega, kappa) generalises	Sleep, ketamine, DOC datasets
R4	Frontal kappa irrelevant	C(frontal) d near 0 across paradigms

Table 5. Falsifiable predictions with specified failure conditions.

10. Limitations

The Overlap-12 cohort is small (N = 7 vs 5). Bootstrap confidence intervals span ~4.5 d-units. The parietal C finding was exploratory. All results derive from one paradigm (propofol sedation). Level 1 ontological claims are framework commitments, not empirically tested here. Transfer entropy parameters (k=3, 6 bins) were not subjected to sensitivity analysis. Replication with larger samples, diverse paradigms, and parameter robustness checks is essential.

11. Conclusion

Consciousness is best described as:

A high-differentiation resonance regime within a universal dynamical field, stabilised by regional feedback rather than global integration.

The ego is not the origin of awareness but a structured expression within this regime — a filter that cannot observe the field conditions that generate it. The strong product hypothesis is falsified. The revised model generates four falsifiable predictions (R1-R4) testable on independent datasets.

Consciousness is becoming a physics problem, not a metaphysical one.

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